Extended Project Analysis

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Data Management II

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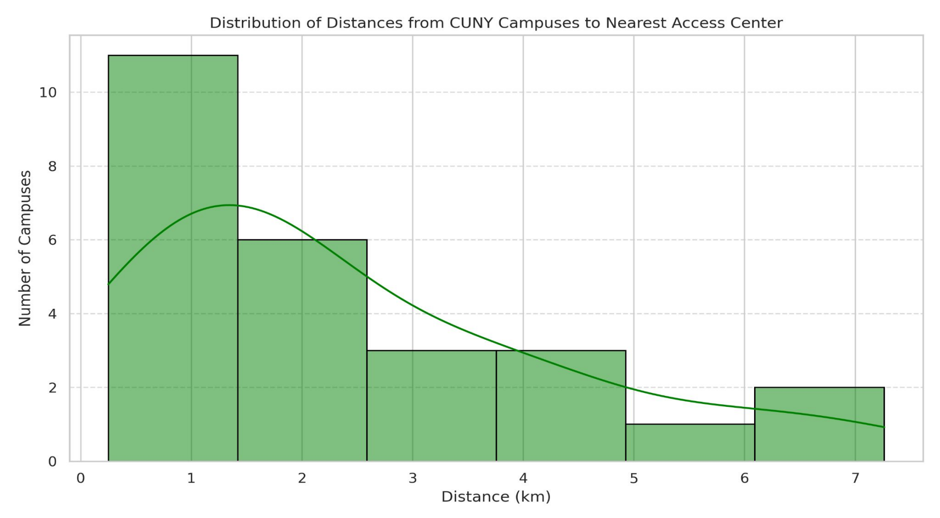
Accessibility of NYC Benefits Access Centers Near CUNY Campuses

Original Project overview

- CUNY campus locations
- NYC Benefits Access Centers
- Goal: evaluate geographic accessibility
- Method: distance-based analysis

Key Findings from Original Analysis



1. Most campuses within 1–2 km of a center

2. Some campuses are significantly farther away

3. Geographic access varies by physical location

4. A few campuses would suggest an urban replaning for easier access to benefit centers.

Why Extend the Project?

- **Distance ≠ service availability**
- **Service descriptions contain unstructured data**
- **Borough-level patterns not previously analyzed**

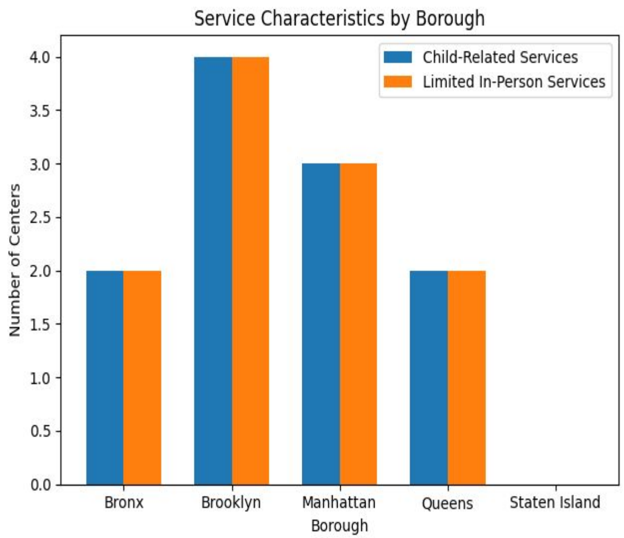
While distance is important, it does not explain whether all access centers provide the same types of services. This raised new questions about service availability and operational differences, which motivated the extension of the project to analyze service characteristics and borough-level patterns in the data set.

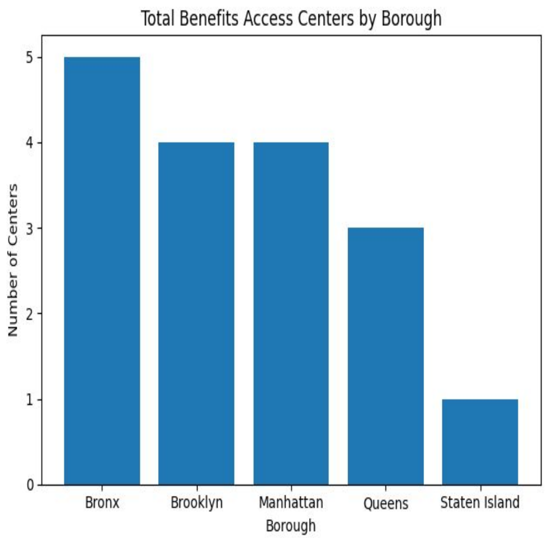
New Techniques Applied:

- Regular Expressions (text feature extraction)
- Pivot Tables (borough-level aggregation)
- Visualization (comparative analysis)
- Prompt Engineering (conceptual)

New Findings from Extended Analysis

- Brooklyn & Manhattan: strong service coverage
- Bronx & Queens: mixed availability
- Staten Island: limited service diversity





Conclusions & Key Takeaways

- Accessibility is multi-dimensional
- Service type matters, not just distance
- Text analysis revealed hidden patterns
- Method choices prioritized insight

Next Steps
QA Session

The Slide 9 is made to answer few possible asked question that will be listed on this slide & Slide 10 was made to ask a classmate a couple of questions regardless their presentation.

The Questions under are asked about my project & next slide I will be answering them

Question #1

Why didn't you use an API even though it was in your proposal presentation?

Question #2

How do service characteristics change the interpretation of accessibility compared to distance alone?

Question #3

What explains the differences in service coverage across boroughs?

Slide #9

Anticipated Questions & Answers

Q1:

After explored the possibility of API usage, I noticed that datasets are relatively stable and not all sources supported consistent API access. So Instead of forcing API usage, I focused on techniques that added direct analytical value, such as text analysis and visualization instead of API's.

Q2:

In the original analysis, most campuses appeared well served based on distance alone. However, the extended analysis showed that service availability varies by borough even when access centers are nearby. For example, Brooklyn and Manhattan offer a higher concentration of child-related services, while Staten Island lacks these features entirely. This shows us that proximity does not guarantee comparable access to supporting services.

Q3:

I believe differences are likely reflect how public services are organized at the borough level rather than campus location alone. Boroughs such as Brooklyn and Manhattan have a higher concentration of access centers that provide broader service features, while other boroughs, like Staten Island, have fewer centers overall. This suggests that administrative structure and service demand may influence availability in addition to geography.

Here is the list of Questions I would like my classmate to answer about their presentation.

- What was the biggest data-cleaning challenge you faced during new/extended analysis?
- Which technique from the course that you used added the most insight to your project in your opinion & Why?
- What would you improve with more time on your hands in the project? would you have done something different? or maybe you would add a new feature to you analysis?

Thank you for your time, I hope you enjoyed the presentation.